
ECHO: Diagnosing Spatial Cue Access in Audio-Language Models

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Stereo localization requires an audio-language model—a system that accepts au-
2 dio and answers in language—to recover spatial evidence from left and right sig-
3 nals and use it to infer direction. Traditional end-to-end accuracy reports only
4 whether the final direction is correct, hiding whether a model failed to obtain the
5 spatial cue or failed to use it. We introduce ECHO (Explicit Cue versus Heard
6 Observation), a paired diagnostic framework, and StereoMusicQA, a benchmark
7 built from real instrument recordings rendered at known positions. Together, they
8 test whether a model can use a spatial cue that it fails to obtain through audio.
9 Each of 1,045 test items has two views: *cue-text* provides the measured left-right
10 level difference and conversion equation; *audio-only* provides labeled channels
11 and requires the model to obtain a useful relationship from audio. Because cue-
12 text also supplies the rule, this comparison measures the practical advantage of
13 a written cue and rule, not the effect of one internal layer. For Qwen2-Audio,
14 Gemini Flash-Lite, and GPT-audio, cue-text left/center/right accuracy is 71.9%,
15 93.6%, and 90.4% compared with 28.5%, 36.6%, and 42.0% audio-only. These
16 43.3–57.0 percentage-point gains remain positive when complete source tracks
17 are resampled. Furthermore, a model can identify the correct side yet still pre-
18 dict an inaccurate angle: cue-text within-5° accuracy ranges from 0.4% to 80.7%,
19 and written-calculation faithfulness ranges from 0.4% to 77.8%. ECHO therefore
20 shows why spatial evaluation should separately test evidence access, cue use, and
21 whether a stated calculation follows from the cue.

22 1 Introduction

23 Spatial information matters whenever an audio system must do more than just name a sound. A
24 music assistant cannot reliably follow a set of instructions like "move the violin towards the center",
25 unless it knows the position of the violin. An accessibility system may recognize a car yet fail to say
26 whether it is on the listener's left or right. A useful audio assistant therefore needs both semantic
27 information—*what* produced the sound—and spatial information—*where* the sound is.

28 Stereo audio contains a left channel and a right channel. When a source is placed to one side, the two
29 channels usually differ. In StereoMusicQA, position is represented by a level difference: a leftward
30 source has more amplitude in the left channel, and a rightward source has more amplitude in the
31 right channel. This measured difference is called the inter-channel level difference, or ILD. ILD is
32 a *spatial cue*: a measurable piece of evidence about position. It is reported in decibels (dB); its sign
33 indicates the louder side, and its magnitude can be converted to an angle using a panning equation.
34 Panning is the studio process of placing a mono sound toward the left or right by changing its level
35 in the two channels.

Recent benchmarks show that general-purpose audio-language models often struggle with basic physical properties such as loudness, pitch, and direction even when they can describe higher-level audio content [Gong et al., 2024, Sun et al., 2026]. Purpose-built systems such as BAT and OWL add spatially specialized encoders or supervision [Zheng et al., 2024, Biswas et al., 2026], but a final localization score still does not reveal whether an error began when the model accessed the spatial evidence or when it used that evidence.

When an audio-language model is asked to localize a stereo source, it faces a two-stage problem. First, the model must compare the left and right audio and obtain a useful cue such as ILD. Second, it must use that cue to decide the side or calculate the angle. We call these stages *cue access* and *cue use*. If a model gives the wrong final answer, an ordinary benchmark cannot tell whether it failed at the first stage, the second stage, or both. That distinction matters because the repairs are different: arithmetic training cannot recover evidence that never reaches the language model, while better early audio processing does not guarantee correct reasoning from an available cue. A single score therefore does not show whether to improve audio processing, channel comparison, or later numerical reasoning.

Explicit Cue versus Heard Observation (ECHO) is designed to reveal which stage fails. StereoMusicQA supplies controlled test items built from real instrument recordings placed at known angles with a reversible panning rule. Here, localization means identifying whether a source is left, centered, or right and estimating its angle. We ask:

When this localization task and its correct answer stay fixed, how much does performance change when the spatial cue and conversion rule are written explicitly rather than left for the model to recover and apply from audio?

For a concrete example, consider one violin placed left of center. The cue-text version gives the model mono (single-channel) audio plus a statement such as "the left channel is 6 dB louder" and provides the conversion equation. The audio-only version gives the two rendered channels but does not state their level difference. Both versions ask for the same side and angle, and both have the same correct answer. If a model succeeds only when the 6 dB difference is written down, it has shown some ability to use that evidence under a supplied rule but has not made the same evidence reliably usable from audio. We call this operational within-item contrast the *cue-access gap*.

Across 1,045 matched items, writing the cue and rule explicitly raises left/center/right accuracy by 43.3-57.0 percentage points for all three tested models, while every audio-only estimate remains below the most-common-label baseline. Correct side predictions still do not guarantee accurate angles or faithful written calculations. Together, these results show that one end-to-end score can hide different failures that require different remedies.

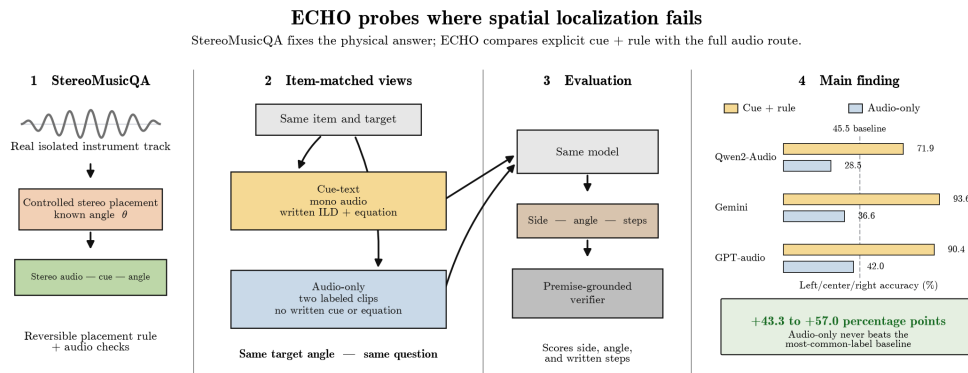


Figure 1: StereoMusicQA places a real instrument recording at a known angle. ECHO asks the same question with the cue and rule written in text or with the relationship available only through audio; the answer, model, and scorer stay fixed. Cue-text improves left/center/right accuracy by 43.3–57.0 percentage points, while every audio-only result remains below the most-common-label baseline.

70 In summary, our contributions are: ¹

- 71 1. **A framework and benchmark.** ECHO compares the same item through the full audio path
72 and best-case explicit cue use. StereoMusicQA uses real recordings, controlled positions,
73 and a reversible rule that makes the cue and answer recomputable.
- 74 2. **Evidence of a cue-access gap.** On the same 1,045 items, three general-purpose audio
75 language models are 43.3–57.0 percentage points more accurate at classifying left, center,
76 or right when the cue and rule are explicit. Confidence intervals obtained by resampling
77 complete source tracks exclude zero for each model.
- 78 3. **Stage-specific scoring.** ECHO separately measures coarse left/center/right decisions, nu-
79 merical angle accuracy, and whether each written calculation follows from the cue stated
80 in the response.

81 2 Related Works

82 **Physical audio evaluation** AV-Odyssey shows that models combining audio and vision can fail
83 basic loudness and pitch comparisons [Gong et al., 2024]. SonicBench evaluates twelve measurable
84 audio properties and reports broad difficulty, including direction [Sun et al., 2026]. Some properties
85 remain predictable from early audio features even when final answers fail. SpeechR finds a related
86 gap in speech: recognizing words does not guarantee reasoning correctly about them [Yang et al.,
87 2025]. ECHO narrows this question to left-right position and changes whether the cue and rule are
88 written explicitly or must be obtained and used through audio.

89 **Purpose-built spatial audio-language systems** BAT combines a specialized spatial audio encoder
90 with a language model [Zheng et al., 2024]; SALM separates semantic and spatial information [Hu
91 et al., 2025]; OWL adds geometric supervision [Biswas et al., 2026]; and Dual-BEATs uses one
92 encoder per channel and reports that training noise can help preserve channel differences through
93 internal normalization [Lin and Huang, 2026]. These systems ask how to build better spatial models.
94 ECHO instead audits where an existing general-purpose system may fail.

95 **Evaluation validity.** Related problems occur beyond audio. Vision-language models can describe
96 scenes while failing simple geometry [Rahmanzadehgervi et al., 2024]; models can score well by
97 learning shortcuts rather than the intended rule [Geirhos et al., 2020]; and plausible step-by-step
98 explanations need not faithfully reflect the basis of an answer [Turpin et al., 2023]. ECHO tests these
99 concerns through item-matched conditions, an off-grid stress test, and a verifier that recomputes
100 arithmetic from the model’s stated cue. Table E1 compares the experimental designs of nearby
101 benchmarks; their scores are not placed on one leaderboard because their tasks and labels differ.

102 3 ECHO and StereoMusicQA

103 ECHO is an evaluation framework for locating where failure may arise inside a spatial-audio
104 question-answering pipeline. StereoMusicQA is the benchmark that makes this comparison pos-
105 sible. The distinction is:

- 106 • **StereoMusicQA creates the problem.** It produces stereo audio with a known source angle,
107 a measurable left-right cue, and a correct answer that can be computed.
- 108 • **ECHO creates the comparison.** It asks the same question once with that cue and its
109 conversion rule stated in text, and once with the relevant relationship available only in the
110 audio channels.

111 The complete evaluation follows five steps. (1) Create a spatial item with known ground truth. (2)
112 Make matched cue-text and audio-only versions. (3) Collect one answer from each model in each
113 condition. (4) Score the reported side and angle, and check any written calculation. (5) Compare the
114 two answers from the same item. The following subsections explain each step and the assumptions
115 behind them.

¹Anonymous code and artifacts include benchmark manifests, prompts, model completions, and evaluation scripts. URMP source audio is not redistributed and must be obtained from its original distribution.

3.1 StereoMusicQA construction

StereoMusicQA begins with the University of Rochester Multi-Modal Music Performance Dataset (URMP) [Li et al., 2019]. URMP contains 149 isolated instrument tracks from 44 classical chamber pieces. An isolated track is a recording of one performer’s part without the other instruments. Each track is mono, meaning that it has one channel and therefore no original left-right position. URMP supplies the real music sound; we create the spatial position ourselves.

We place each mono recording at a chosen horizontal angle and render a new left and right channel. Positive angles mean left, zero means center, and negative angles mean right. The main study uses 1,045 held-out single-source items made from 19 tracks in six pieces. We split by complete musical piece, so a performance used for testing never appears in training.

Each held-out track is rendered at eleven angles from -25° to $+25^\circ$ under an unchanged condition and four common audio transformations. Because these are repeated versions of 19 recordings, Section 3.5 treats them as related observations rather than 1,045 independent sources. Appendix D.1 gives full construction details.

For the main experiment, we use amplitude panning. This is the familiar studio operation that moves a mono signal left or right by changing its level in the two output channels. The panning law uses virtual speaker anchors at $+30^\circ$ on the left and -30° on the right; the tested source angles stay within $\pm 25^\circ$. The interchannel level difference is defined as $ILD = 20 \log_{10}(L/R)$, where L and R are the left- and right-channel amplitudes. We call the horizontal source angle the azimuth and denote it by θ . Thus positive ILD and positive θ indicate left, while negative values indicate right. If $r = L/R$, then

$$r = 10^{ILD/20}, \quad d = \frac{r-1}{r+1}, \quad \theta = \arctan(d \tan 30^\circ).$$

This equation makes the benchmark checkable in both directions. We know the angle used to create the audio, and we can measure the finished left and right channels, recover ILD, and obtain the same angle. Items that fail this check are rejected.

This design is controlled studio panning, not a simulation of natural human hearing, which also depends on the listener’s anatomy, room, frequency, elevation, and distance. We use the simpler case because it gives an exact link among waveform, cue, and answer.

3.2 Two conditions for one localization problem

Each item is presented in two matched ways:

- **Cue-text condition.** The model receives mono audio, so it still hears the musical content, but the prompt explicitly states the measured ILD, identifies the louder channel, and gives the panning equation. The model must use the supplied cue by following the provided rule.
- **Audio-only condition.** The model receives the left and right channels as two separately labeled audio clips. The prompt does not state ILD or the panning equation. The model must compare the clips and obtain a useful spatial relationship through its audio interface.

The source rendering, planted position, question, correct answer, model, and scoring rule remain fixed. The information supplied to the model is not otherwise identical: cue-text provides both the measured cue and the conversion rule, whereas audio-only provides neither. The paired gap therefore measures the practical advantage of explicit symbolic access over the full audio route. It does not cleanly separate extracting the cue from knowing or applying the conversion rule. An audio-plus-rule condition would be needed to isolate that distinction more tightly.

There is also an important delivery limitation: Qwen2-Audio’s audio input component accepts mono audio, so we cannot pass one native stereo file consistently across all systems. Separate labeled clips preserve left and right identity, but the model must still compare two audio sequences. Thus ECHO measures the complete practical access path, including audio ingestion, cross-channel comparison, alignment with the language model, and answer generation. This comparison alone cannot identify one responsible internal layer.

163 3.3 Models

164 We test Qwen2-Audio-7b-Instruct locally [Chu et al., 2024] and Gemini Flash-Lite and GPT-audio
165 through their hosted application programming interfaces (APIs). Each model receives all 1,045
166 items in both conditions, with instructions but no solved examples, and produces one answer per
167 item. Only Qwen2-Audio exposes the internal features needed for probing and fine-tuning. Ap-
168 pendix D records exact settings and the limitations of changing hosted services.

169 3.4 Scoring and checking written calculations

170 We score three outcomes. **Left/center/right accuracy** is the percentage assigned the correct position
171 label. Table A1 also reports class-balanced measures so a frequent default answer is not rewarded.
172 **Angle accuracy** asks whether the prediction is within 5° of the planted angle; mean absolute error
173 reports the average error size. **Stated-derivation faithfulness** asks whether the visible calculation
174 follows mathematically from the cue in the response.

175 Unanswered items count as wrong. In cue-text answers, the final signed angle determines side
176 because the prompt defines positive as left and negative as right. In audio-only answers, we score
177 the explicit word "left", "center", or "right" because that prompt does not introduce a numeric sign
178 convention. Every model uses the same extraction rule within a condition.

179 We implement a deterministic *verifier*: a program that parses the model’s written steps and recom-
180 puts them from the stated premise rather than checking only the final answer. If the model writes
181 $r = 10^{x/20} = y$, it recomputes y from the stated x before checking the later angle. This catches
182 a response that copies the cue but silently calculates with a memorized value. We call the check
183 *premise-grounded* because it begins with the premise stated in the answer. Appendix B documents
184 the off-grid audit that motivated this stricter check: a response can resemble a solved equation using
185 a value that does not follow from its own first line.

186 3.5 Uncertainty and dependence

187 The 1,045 items are not 1,045 unrelated recordings. They reuse 19 source tracks at different angles
188 and with different processing. Treating every render as independent would make the result look
189 more certain than it is. We therefore resample complete tracks, rather than individual items, 20,000
190 times while keeping each item’s two conditions paired. The middle 95% of the resulting differences
191 form the confidence interval. A stricter check groups the tracks by their six musical pieces; all paired
192 intervals still exclude zero. Appendix D.3 gives the procedure and cautions that six pieces provide
193 limited precision.

194 4 Results

195 4.1 Explicit cues and rules reveal a large performance gap

196 As Figure 2 shows, left/center/right accuracy is higher for every model when the level difference and
197 conversion rule are written in the prompt than when the model must obtain and use a relationship
198 from audio.

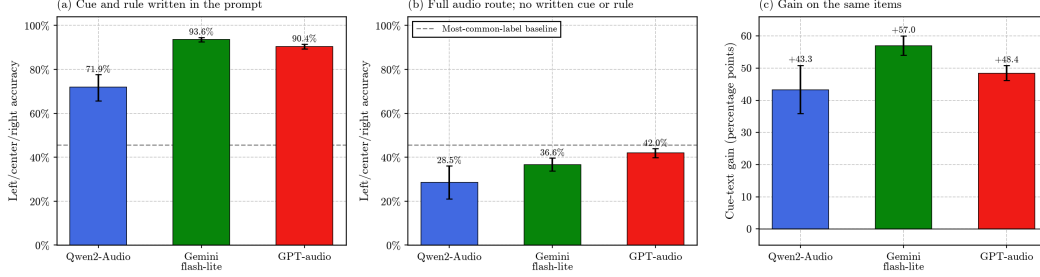


Figure 2: The main paired result. Panels (a) and (b) show left/center/right accuracy: the percentage of items assigned the correct one of those three labels. Panel (a) supplies the level cue and conversion rule in text; panel (b) requires the model to obtain and use a relationship from audio. The dashed line marks 45.5%, the score from always choosing the most common test label. Panel (c) subtracts audio-only accuracy from cue-text accuracy for each model. Error bars are 95% confidence intervals obtained by resampling the 19 held-out source tracks 20,000 times while keeping each item’s two conditions paired.

Qwen2-Audio improves from 28.5% to 71.9%, Gemini Flash-Lite from 36.6% to 93.6%, and GPT-audio from 42.0% to 90.4%. These are increases of 43.3, 57.0, and 48.4 percentage points on the same items. Every 95% interval for the paired increase stays above zero. None of the audio-only estimates, or their intervals, reaches the 45.5% score of a trivial system that always chooses the most common label.

The errors do not reflect one common physical interpretation. Instead, each model tends to fall back on a preferred answer: Qwen2-Audio often says right, while Gemini and GPT-audio often say left. This pattern is consistent with answer bias rather than reliable use of the channel relationship.

The right label is somewhat more common than center, so raw accuracy can reward a default answer. Table A1 therefore reports balanced accuracy and macro-F1, which give equal importance to left, center, and right.

Audio-only balanced accuracy stays near 32% for all three models, close to the 33.3% expected from equal performance across three classes. The conclusion also survives the stricter analysis that groups tracks by their six musical pieces: the paired intervals remain above zero for all models. Because six pieces are a small number of groups, we treat this as a sensitivity check rather than the main interval.

This result does not mean the audio contains no information, and it does not identify one broken layer. It means that the complete audio route does not make the relevant left-right evidence reliably usable by the answer-generating system, even though the same system performs much better once that evidence and its conversion rule are written explicitly.

4.2 Correct side predictions do not guarantee accurate angles

Choosing left or right requires only the sign of the level difference. Calculating an angle requires using its numerical size correctly. Figure 3 summarizes this separation; Table A1 reports the complete metrics.

Qwen2-Audio reaches 71.9% cue-text left/center/right accuracy but only 0.4% within-5° angle accuracy. It usually identifies the louder side without carrying out the supplied conversion correctly. Gemini and GPT-audio calculate the angles more successfully, at 44.9% and 80.7% within 5°, but their written calculations are not always faithful to the stated cue. Premise-grounded verifier faithfulness is 0.4%, 42.4%, and 77.8% for Qwen2-Audio, Gemini, and GPT-audio, respectively.

A common error is *boundary substitution*. The prompt explains that the virtual speakers are at -30° and $+30^\circ$. Instead of evaluating the equation for the current item, the model sometimes repeats 30 degrees as if it were the answer. This occurs in nearly every answered Qwen2-Audio item, 19.7% of Gemini’s answered items, and 1.2% of GPT-audio’s.

Qwen2-Audio	0.4	71.9	0.4	28.5
Gemini Flash-Lite	44.9	93.6	42.4	36.6
GPT-audio	80.7	90.4	77.8	42.0
	Cue-text within 5°	Cue-text left/center/right	Written calculation faithfulness	Audio-only left/center/right

Figure 3: Correct side does not guarantee an accurate angle or a faithful calculation. Values are percentages on the same 1,045 cue-text items, except the final column, which reports audio-only left/center/right accuracy. Written-calculation faithfulness measures whether the displayed steps follow from the cue stated in the response.

Qwen2-Audio’s mean error can exceed the benchmark’s 30-degree limit because we score the angle exactly as generated. We do not silently clip an answer such as 90 degrees back into the allowed range.

Two supporting analyses are reported outside the main results. Appendix B documents the off-grid fine-tuning audit that motivated premise-grounded verification. Appendix D.5 reports a preliminary Qwen2-Audio encoder probe; the probe supplies its own left-right subtraction, so it is not evidence that the complete model performs that comparison.

5 Discussion

What ECHO establishes. On the same items and models, performance changes sharply when the cue and rule are written down. This shows a practical difference between best-case explicit cue use and the full audio route; it does not isolate one mental ability or encoder layer. "Cue access" includes channel encoding, comparison, audio–language alignment, answer generation. The distinction still matters because arithmetic training cannot recover a cue that never becomes usable, while better audio processing does not ensure correct calculation.

Cue-text success also has a narrow meaning: the model used explicit information under this prompt, not that it possesses general spatial reasoning. Likewise, our verifier checks whether the visible calculation follows from the stated cue. A convincing derivation can still contain arithmetic that does not follow from the evidence written in its first line.

Case study: locating a failure after the cue. Consider one held-out off-grid item whose true position is 21.8° to the right and whose measured ILD is -14.8 dB. The fine-tuned Qwen2-Audio response correctly repeats that cue and identifies the right side, but then writes $r = 10^{-14.8/20} = 0.139$ and concludes -23.6° . ECHO’s verifier recomputes the displayed step and finds that the ratio should be approximately 0.182, not 0.139. The written value 0.139 instead corresponds to a nearby -17.2 dB training cue whose stored answer is -23.6° . This example locates the failure after the premise is available: the model copies the new cue but substitutes a memorized cue during calculation. The same pattern occurs in 189 of 190 original off-grid responses. A final-answer metric alone would report only a 1.8° error; the premise-grounded check reveals the shortcut that produced it.

260 **Implications.** The Qwen probe suggests that jointly comparing channels may be more useful than
261 simply unfreezing an entire encoder. For evaluation, file count is not problem diversity: benchmarks
262 should report distinct cue–answer relationships, test values between training points, and verify each
263 written step from the model’s stated premise. Preliminary dithering experiments perturb the failure
264 but collapse to a few answer templates; Appendix C reports them as future-work evidence, not a
265 solution.

266 **Broader impact.** More diagnostic spatial evaluation could improve the reliability of music-
267 production tools, accessibility systems, and other audio assistants by showing whether a failure
268 comes from unavailable evidence or incorrect use of that evidence. The main risk is overgeneral-
269 ization: success on controlled amplitude panning does not establish reliable localization in rooms,
270 natural binaural hearing, or safety-critical settings. We mitigate this risk by releasing the verifier
271 and benchmark audit, reporting negative results, and limiting our claims to the tested cue, delivery
272 protocol, and model families.

273 6 Limitations

274 While ECHO reveals a consistent operational gap on the tested systems, we note several limitations.
275 The positions use controlled studio panning, not natural hearing. Cue-text provides both the cue
276 and rule, so an audio-plus-rule condition is still needed to separate cue extraction from knowledge
277 of the rule. Left and right are delivered as two labeled clips rather than one native stereo file. The
278 study covers 19 tracks from six pieces and only three model families; the undisclosed sizes of
279 Gemini Flash-Lite and GPT-audio prevent a meaningful analysis of scaling. Hosted systems may
280 also change. The two conditions require different fixed parsing rules, and hosted models are sampled
281 once per item.

282 7 Conclusion

283 ECHO treats localization as a sequence of steps rather than one final score: audio must contain a
284 cue, the model must make that cue usable, the cue must be converted into an answer, and any written
285 calculation should follow from the evidence provided. StereoMusicQA makes each stage checkable
286 by connecting a real instrument recording, a measured left-right difference, and a known source
287 angle.

288 Across Qwen2-Audio, Gemini Flash-Lite, and GPT-audio, writing the cue and conversion rule in the
289 prompt improves left/center/right accuracy by 43.3–57.0 percentage points, while audio-only accu-
290 racy remains below the score from always choosing the most common label. The models therefore
291 perform substantially better under best-case explicit cue use than through the complete audio route.
292 Exact audio calculation and written-calculation faithfulness remain separate weaknesses. These find-
293 ings show why spatial evaluation should report where the pipeline fails, not only whether the last
294 answer is correct. More broadly, evaluation can overstate a model’s ability whenever it confuses
295 access to evidence with reasoning from that evidence, or mistakes a convincing written derivation
296 for a faithful one.

297 References

- 298 Subrata Biswas, Mohammad Nur Hossain Khan, and Bashima Islam. OWL: Geometry-aware spatial
299 reasoning for audio large language models. In *International Conference on Learning Representa-*
300 *tions*, 2026.
- 301 Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, Yichong Leng, Yuanjun Lv,
302 Jinzheng He, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen2-Audio technical report. *arXiv*
303 *preprint arXiv:2407.10759*, 2024. doi: 10.48550/arXiv.2407.10759.
- 304 Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel,
305 Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature*
306 *Machine Intelligence*, 2(11):665–673, 2020. doi: 10.1038/s42256-020-00257-z.

- 307 Kaixiong Gong, Kaituo Feng, Bohao Li, Yibing Wang, Mofan Cheng, Shijia Yang, Jiaming Han,
308 Benyou Wang, Yutong Bai, Zhuoran Yang, and Xiangyu Yue. AV-Odyssey Bench: Can your
309 multimodal LLMs really understand audio-visual information? *arXiv preprint arXiv:2412.02611*,
310 2024. doi: 10.48550/arXiv.2412.02611.
- 311 Jinbo Hu, Yin Cao, Ming Wu, Zhenbo Luo, and Jun Yang. SALM: Spatial audio language model
312 with structured embeddings for understanding and editing. *arXiv preprint arXiv:2507.16724*,
313 2025. doi: 10.48550/arXiv.2507.16724.
- 314 Bochen Li, Xinzhaio Liu, Karthik Dinesh, Zhiyao Duan, and Gaurav Sharma. Creating a multitrack
315 classical music performance dataset for multimodal music analysis: Challenges, insights, and
316 applications. *IEEE Transactions on Multimedia*, 21(2):522–535, 2019. doi: 10.1109/TMM.2018.
317 2856090.
- 318 Shuo-Chun Lin and Hen-Hsen Huang. Dual-BEATs: Unlocking zero-shot stereo audio perception
319 in audio large language models via dithering. *arXiv preprint arXiv:2607.08800*, 2026. doi: 10.
320 48550/arXiv.2607.08800.
- 321 Pooyan Rahmazadehgervi, Logan Bolton, Mohammad Reza Taesiri, and Anh Totti Nguyen. Vision
322 language models are blind. In *Proceedings of the Asian Conference on Computer Vision*, pages
323 18–34, December 2024.
- 324 Yirong Sun, Yanjun Chen, Xin Qiu, Gang Zhang, Hongyu Chen, Daokuan Wu, Chengming Li,
325 Min Yang, Dawei Zhu, Wei Zhang, and Xiaoyu Shen. SonicBench: Dissecting the physical
326 perception bottleneck in large audio language models. *arXiv preprint arXiv:2601.11039*, 2026.
327 doi: 10.48550/arXiv.2601.11039.
- 328 Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don’t always
329 say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in
330 Neural Information Processing Systems*, volume 36, pages 74952–74965, 2023. doi: 10.52202/
331 075280-3275.
- 332 Wanqi Yang, Yanda Li, Yunchao Wei, Meng Fang, and Ling Chen. SpeechR: A benchmark for
333 speech reasoning in large audio-language models. *arXiv preprint arXiv:2508.02018*, 2025. doi:
334 10.48550/arXiv.2508.02018.
- 335 Zhisheng Zheng, Puyuan Peng, Ziyang Ma, Xie Chen, Eunsol Choi, and David Harwath. BAT:
336 Learning to reason about spatial sounds with large language models. In *Proceedings of the 41st
337 International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning
338 Research*, pages 61454–61469. PMLR, 2024.

339 A Complete zero-shot results

340 Balanced accuracy averages the fraction correctly identified within each of the three position labels.
341 Precision asks how often a predicted label is correct, recall asks how many true items with that label
342 are found, and F1 combines the two. Macro-F1 computes F1 separately for left, center, and right
343 and averages the three. Coverage is the fraction of responses containing an extractable angle.

Table A1: Complete zero-shot results for the three evaluated systems.

Metric	Qwen2-Audio	Gemini Flash-Lite	GPT-audio
Cue-text within-5°	0.4%	44.9%	80.7%
Cue-text mean absolute angle error	74.27°	7.49°	5.44°
Cue-text left/center/right accuracy	71.9%	93.6%	90.4%
Cue-text verifier faithfulness	0.4%	42.4%	77.8%
Audio-only left/center/right accuracy	28.5%	36.6%	42.0%
Cue-text balanced accuracy	53.0%	83.8%	82.6%
Audio-only balanced accuracy	31.9%	32.7%	31.4%
Cue-text macro-F1	51.4%	88.7%	86.4%
Audio-only macro-F1	26.0%	31.2%	28.0%
Cue-text coverage	97.6%	96.8%	99.8%
Boundary-substitution rate among answered items	~ 100%	19.7%	1.2%

Table A2: Exact left/center/right results for the two item-matched ECHO conditions. Brackets show 95% confidence intervals obtained by resampling complete source tracks while preserving each cue text/audio-only pair; pp means percentage points.

Model	Cue-text accuracy	Audio-only accuracy	Paired gap
Qwen2-Audio	71.9% [65.6, 77.6]	28.5% [21.0, 36.1]	+43.3 pp [35.9, 50.9]
Gemini Flash-Lite	93.6% [92.5, 94.6]	36.6% [33.7, 39.6]	+57.0 pp [54.0, 60.0]
GPT-audio	90.4% [89.4, 91.5]	42.0% [39.9, 44.0]	+48.4 pp [46.2, 50.8]

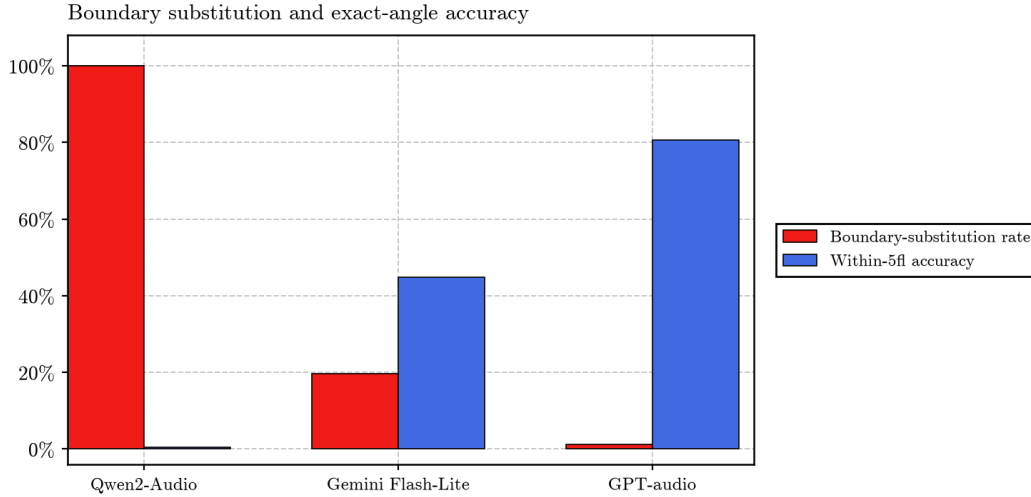


Figure A1: Descriptive results for the three evaluated systems. Bars are not connected because three model endpoints do not establish a scaling trend.

B Off-grid audit and bounded repair

We originally fine-tuned Qwen2-Audio on cue-text examples. Its within-5° accuracy increased from 0.4% to 100%, but the training and standard test sets displayed the same 21 cue-answer pairs at different recordings. We therefore created 190 off-grid items at ten new angles between the trained values. Every prediction landed on one of the old 21 angles. In 189 of 190 responses, the model first

349 copied the new cue and then calculated with a different value from the training grid. The answer
 350 looked like a solved equation, but the arithmetic did not follow from its own premise.

351 This audit motivated the premise-grounded verifier used in the main study. The original verifier
 352 checked intermediate values against stored labels; the corrected version recomputes every displayed
 353 step from the cue written in the response. Only 0.5% of the original off-grid derivations pass that
 354 stricter check. This appendix records the failed fine-tuning attempt as a benchmark audit, not as a
 355 general claim about audio-language models.

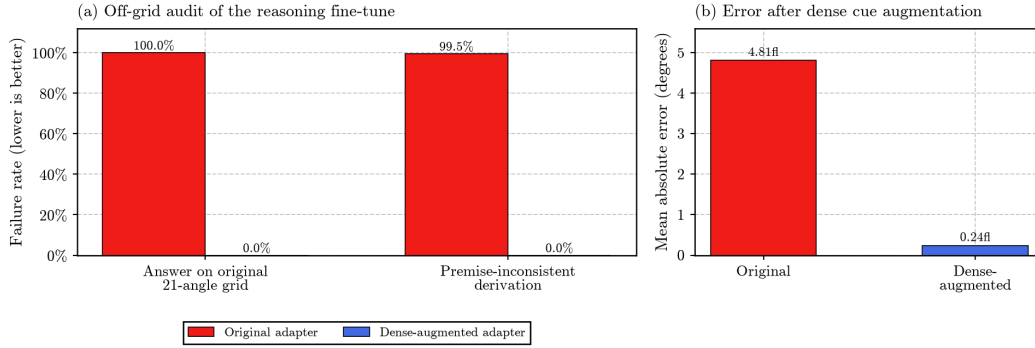


Figure B1: The original fine-tuned model’s off-grid outputs expose the 21-row lookup. “Off-grid” means that the test cue lies between values used during training. Every answer lands on the old training grid, and 99.5% contain arithmetic that does not follow from the cue stated in the response. Adding many intermediate cue values to training removes those specific indicators and reduces mean absolute error (MAE) from 4.81° to 0.24° .

356 The dense-augmentation run adds 3,000 continuously sampled cue-answer examples, producing
 357 606 displayed ILDs and 430 displayed azimuths. The ten off-grid test angles are excluded during
 358 sampling. On the original 1,045-item test set, the fine-tuned model retains 100% within- 5° accuracy
 359 and reaches 100% premise-grounded faithfulness. On the 190 off-grid items, it reaches 100% within-
 360 5° , 0.24° MAE, and 100% verifier faithfulness.

361 These results eliminate the specific 21-row failure. They do not prove that the model executes the
 362 symbolic law. All 190 repaired predictions fall within 0.2° of some dense training azimuth, so
 363 accurate interpolation remains a complete alternative explanation. The repair is therefore reported
 364 as benchmark hardening, not as proof of human-like computation.

365 C Preliminary dithering experiment

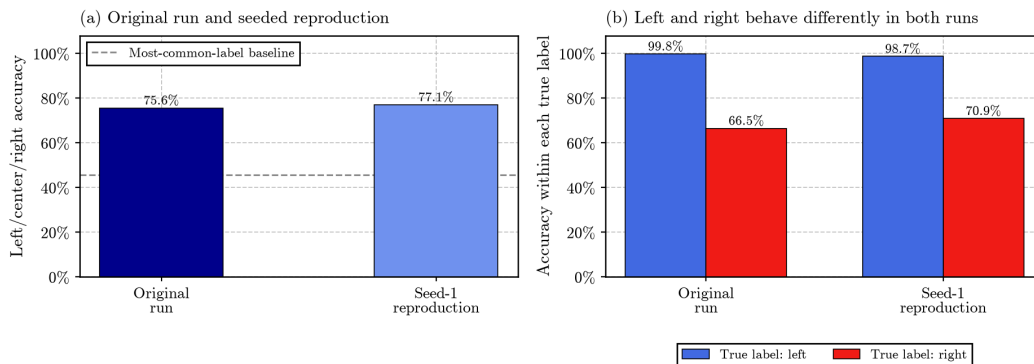


Figure C1: Training-time dithering—adding small random noise to each audio channel during training—raises Qwen2-Audio left/center/right accuracy to 75.6% in the original run and 77.1% in a reproduction with training-noise seed 1. Both fine-tuned models are evaluated with the same noise sequence, identified by seed 101. The output remains severely template-collapsed: it reuses only a few answer patterns and never identifies the center correctly.

Table C1: Left/center/right accuracy for the original dithering run and its seeded reproduction.

Metric	Original run	Seed-1 reproduction
Left/center/right accuracy	75.6%	77.1%
True-left correct	99.8%	98.7%
True-right correct	66.5%	70.9%
True-center correct	0.0%	0.0%

366 Inference-only dithering does not produce a reliable gain. Training-time dithering does, but the
 367 model emits only a few fixed sentence templates and does not recover exact localization. Fully
 368 unfreezing the encoder regresses to 45.5% after one epoch and 46.4% after three epochs, both ex-
 369 plained by answer collapse rather than true discrimination. These experiments motivate future work
 370 on joint channel representations or the audio-language projector.

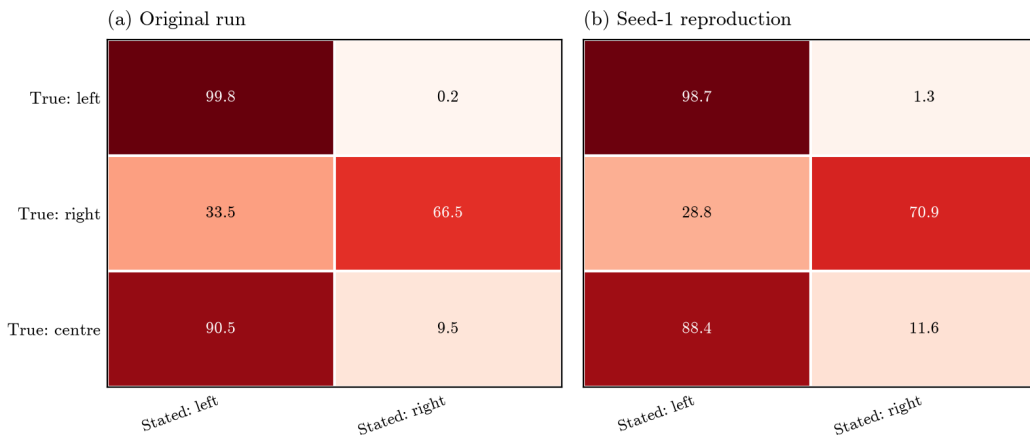


Figure C2: Dithering confusion matrices. Each row is the true position label, and each column is the label predicted by the model.

D Reproducibility notes

D.1 Benchmark construction

We sort the 44 URMP piece identifiers and shuffle them once with Python’s Mersenne Twister using seed 2026. A 70/15/15 piece-level split produces 31 training pieces, seven development pieces, and six test pieces; no isolated source track (or *stem*) from a piece crosses splits. The inventory contains 149 mono instrument stems. For each stem at least six seconds long, we derive a stem-specific 64-bit seed from the first eight bytes of SHA-256(“2026:”) and use that seed to draw one uniform six-second excerpt offset. The same excerpt is reused for all angles and processing conditions, so comparisons do not confound position with musical content.

Each excerpt is resampled to 48 kHz (48,000 audio samples per second) and rendered at -25° , -20° , -15° , -10° , -5° , 0° , 5° , 10° , 15° , 20° , and 25° . The dry render, meaning no additional processing, uses the tangent amplitude-panning rule with virtual speakers at $\pm 30^\circ$. Four additional renders apply one transformation after panning: 4:1 dynamic-range compression above -20 dB, a $+6$ dB equalizer that changes frequency balance around 1 kHz, stereo widening that separately scales the shared and differing parts of the two channels (mid/side widening) with width 1.2, or a 300 microsecond delay on the right channel. This gives 8,195 candidate renders. A candidate is rejected if it is non-stereo, contains invalid numerical samples, is quieter than -45 dB root-mean-square (RMS) level—a measure of average signal level—has peaks above 0.99, or fails to recover the processing-adjusted azimuth within 5° . We additionally require the expected inter-channel delay within 30 microseconds, the expected regime label, and channel coherence—a measure of similarity between the two channel waveforms—of at least 0.90 whenever the processing should preserve a narrow spatial source. These checks retain 7,718 items: 5,332 training, 1,341 development, and 1,045 test items. Audio is stored as 24-bit pulse-code modulated (PCM) WAV files, a standard uncompressed digital-audio format; public test identifiers conceal stem, offset, and angle, while those fields are retained in the private labels.

D.2 Zero-shot inference

Every model receives the same held-out items and one completion per condition; we do not ask the same question several times and vote among the answers, a procedure often called self-consistency sampling. For cue-text evaluation, ILD is measured on the 48 kHz render before the audio is down-mixed to mono, and the prompt supplies that measurement and the panning law. For audio-only evaluation, the left and right waveforms are supplied as separately identified audio inputs and the prompt contains no measured cue. Qwen2-Audio resamples each input independently to 16 kHz, uses the bfloat16 numerical format, and always selects its highest-scoring next token (greedy decoding) with a limit of 200 new tokens. Gemini Flash-Lite and GPT-audio receive 48 kHz 16-bit PCM WAV data through their provider APIs. Those hosted calls used the providers’ default generation settings, which did not expose a fixed seed in our evaluation path. Their reported values are therefore one-answer measurements of those hosted service versions, not estimates averaged over repeated random seeds. Failed API calls were retried up to three times; completed item identifiers were checkpointed so a resumed run did not generate a second answer for an already completed item.

D.3 Statistical analysis

The headline confidence intervals use the paired cluster bootstrap described in Section 3.5: 20,000 resamples of the 19 complete held-out stem clusters with replacement, seed 20,260,814, and percentile endpoints at 2.5% and 97.5%. Pairing is preserved both across conditions and across all transformations of a stem. No item-level independence assumption is used, and no correction for multiple comparisons is applied. The majority baseline is deterministic: it always predicts right, one of two tied most-frequent test labels, and is correct for 475/1,045 items (45.5%); always-left has the same accuracy.

As a stricter dependence sensitivity analysis, we repeat the paired bootstrap while clustering by the six held-out URMP pieces. The paired-gap intervals are [37.3, 50.2], [55.5, 59.4], and [46.3, 50.5] percentage points for Qwen2-Audio, Gemini Flash-Lite, and GPT-audio, respectively. All exclude zero, but the small number of piece clusters makes these intervals secondary to the stem-cluster

analysis. We also compute balanced accuracy and macro-F1 directly from the three-class confusion matrices; these are point estimates, not additional hypothesis tests.

D.4 Qwen2-Audio fine-tuning and off-grid audit

All Qwen fine-tunes use quantized low-rank adaptation (QLoRA), which trains small added weight matrices while keeping the 4-bit NormalFloat (NF4)-quantized base model fixed. Computation uses bfloat16 and gradient checkpointing, which saves memory by recomputing some intermediate values, QLoRA rank 16, scale 32, and dropout 0.05, meaning that 5% of the adapter’s units are randomly disabled during each training step. AdamW uses a learning rate of 2×10^{-4} , one example per forward pass, and eight-step gradient accumulation, meaning that eight examples contribute before each weight update. Unless an experiment states otherwise, training makes one pass over the 5,332 training items. Prompt tokens are masked, so the loss is applied only to the target answer. The original suffix-based LoRA target list adapts the language model’s query, key, value, output, gate, up, and down projections; because Qwen2-Audio reuses the query/key/value names in its audio tower, it also adapts those three projections in all 32 encoder layers. The audio-to-language projector, which connects audio features to the language model, and the remaining encoder weights stay frozen in that configuration.

The dense repair appends 3,000 cue-answer pairs generated with seed 2026. Azimuths are sampled uniformly from $[-29.5^\circ, 29.5^\circ]$, rejecting values within 0.2° of any of the ten off-grid evaluation angles. Each sampled azimuth is converted to ILD by the exact inverse panning law, and its target derivation is computed forward from the displayed cue. These examples reuse real training audio only as the required audio placeholder; the cue-text condition is deliberately solved from the supplied symbolic cue. The repaired QLoRA adapter is trained for one pass over 5,332 real plus 3,000 synthetic examples and is evaluated both on the original 1,045 items and on 190 items constructed at ten withheld angles: $\pm 2.5^\circ$, $\pm 8.0^\circ$, $\pm 13.5^\circ$, $\pm 16.4^\circ$, and $\pm 21.8^\circ$.

Not every training source of randomness was fixed in the historical runs. The original reasoning adapter and dense-repair adapter are single optimization runs without an explicitly recorded PyTorch/CUDA seed. Likewise, the first dithering run drew training noise from an unseeded generator. The dithering reproduction fixes training-noise and Torch seed 1; both the original and reproduced adapters are then evaluated on the same independent Gaussian noise sequence with seed 101. Dithering adds independent Gaussian noise drawn from the standard normal distribution, $N(0, 1)$, to every sample of each channel at amplitude 0.05. Thus the 75.6%/77.1% agreement is a seeded reproduction of the intervention’s qualitative effect, but the other fine-tune numbers should not be presented as multi-seed means.

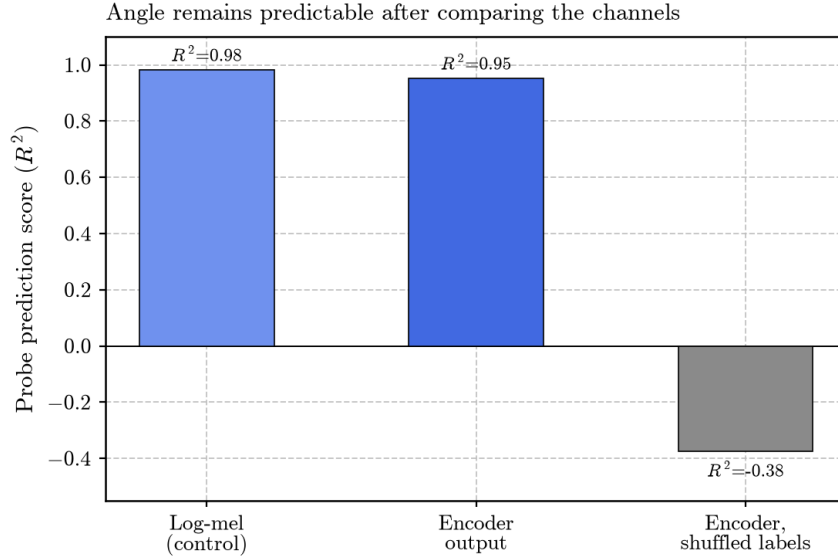


Figure D1: A simple diagnostic predictor recovers source angle when we explicitly subtract right-channel features from left-channel features. Prediction reaches $R^2 = 0.98$ from standard log-mel features and $R^2 = 0.95$ from frozen Qwen2-Audio encoder features. An R^2 near 1 indicates strong prediction. Shuffling the angle labels provides a negative control and produces $R^2 = -0.38$.

457 The encoder probe uses seed 2026. From the held-out split, it selects ten six-second excerpts; for
 458 each selected stem it draws 20 candidate offsets and keeps the one with the highest RMS. Each
 459 excerpt is rendered at the eleven main benchmark angles, producing 110 observations. Left and right
 460 channels are encoded separately, feature vectors from valid time frames are averaged (mean-pooled),
 461 and the right feature vector is subtracted from the left. A ridge regressor—a linear model with an L_2
 462 penalty—is evaluated by leave-one-excerpt-out cross-validation. The procedure repeatedly trains on
 463 nine excerpts and tests on the remaining one, so all eleven angles from the test excerpt are excluded
 464 from its training fold. The shuffled control permutes angle labels within each excerpt with seed 2027
 465 while preserving features, groups, and fold sizes.

466 During the off-grid audit, we found that the original verifier compared claims with stored ground
 467 truth but did not recompute an intermediate value from the premise written in the model’s own
 468 derivation. The corrected verifier parses expressions of the form $r = 10^{x/20} = y$ and checks y
 469 against the stated x before checking the final angle. We also found that an earlier target generator
 470 displayed ILD rounded to one decimal while computing an intermediate value from the unrounded
 471 ILD. The discrepancy was at most 0.05 dB and did not alter a reported aggregate, but all current
 472 targets compute downstream quantities from the displayed premise. Benchmark manifests, prompts,
 473 raw model completions, private scoring labels, rejection records, and experiment configurations
 474 are available in the anonymous artifact repository; URMP source audio must be obtained from its
 475 original distribution.

E Design comparison with related benchmarks

Table E1: Design comparison with adjacent physical-audio and spatial-audio resources. The table distinguishes evaluation benchmarks from systems that introduce specialized spatial training.

Resource	Primary question	Spatial evidence and labels	Experimental contrast	Audit
AV-Odyssey [Gong et al., 2024]	Can multimodal models combine audio and visual clues, including basic loudness and pitch?	Carefully constructed audio-visual multiple-choice problems; not a spatial-localization benchmark	Audio-visual reasoning across 4,555 problems	Objective choices; a matched spatial cue intervention is not reported
SonicBench [Sun et al., 2026]	How broadly do models perceive physical audio attributes?	Controlled stimuli for 12 attributes; direction uses coarse front-versus-back sectors	Recognition versus comparison across 2,400 items	Human and random baselines plus encoder probes; a matched explicit-cue/audio comparison is not reported
SpatialSoundQA / BAT [Zheng et al., 2024]	Can a purpose-built model perceive and reason about binaural scenes?	AudioSet sources rendered with SoundSpaces 2.0; localization, distance, and spatial-relation QA	Specialized Spatial-AST-LLM training versus prior systems	Evaluates spatial QA performance; a premise-grounded arithmetic verifier is not reported
BiDepth / OWL [Biswas et al., 2026]	Can geometric supervision improve direction, distance, and spatial reasoning?	Binaural audio paired during training with panoramic depth and room impulse responses; o'clock-level azimuth and direction-of-arrival (DoA) labels	Geometry-aware SAGE/OWL versus BAT on BiDepth and SpatialSoundQA	Spatially grounded step-by-step reasoning supervision; matched cue-access and off-grid arithmetic audits are not reported
StereoMusicQA / ECHO (ours)	Can a model use a spatial cue that it fails to access through audio?	Real URMP stems rendered at invertible planted horizontal angles; left/center/right label, continuous angle, and recomputable ILD	The same item under explicit cue-text versus audio-only access	Premise-grounded derivation verification and withheld off-grid cues